Document Search and Summarization Using LLMs — Project Report

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This report covers a system that searches 2,089 BBC news articles with hybrid search (BM25 + sentence embeddings) and summarizes the top results with an LLM. Hybrid search found the correct article in the top 5 for all 50 test queries; summaries were highly relevant (4.9 / 5) but accuracy (3.6 / 5) was the weakest area.

1. Data pre-processing

Dataset: BBC News Summary (Kaggle), 2,225 news articles from 2004–2005 in five categories (business, entertainment, politics, sport, tech). It was chosen because every article comes with a human-written reference summary, which allows automatic scoring of summaries.

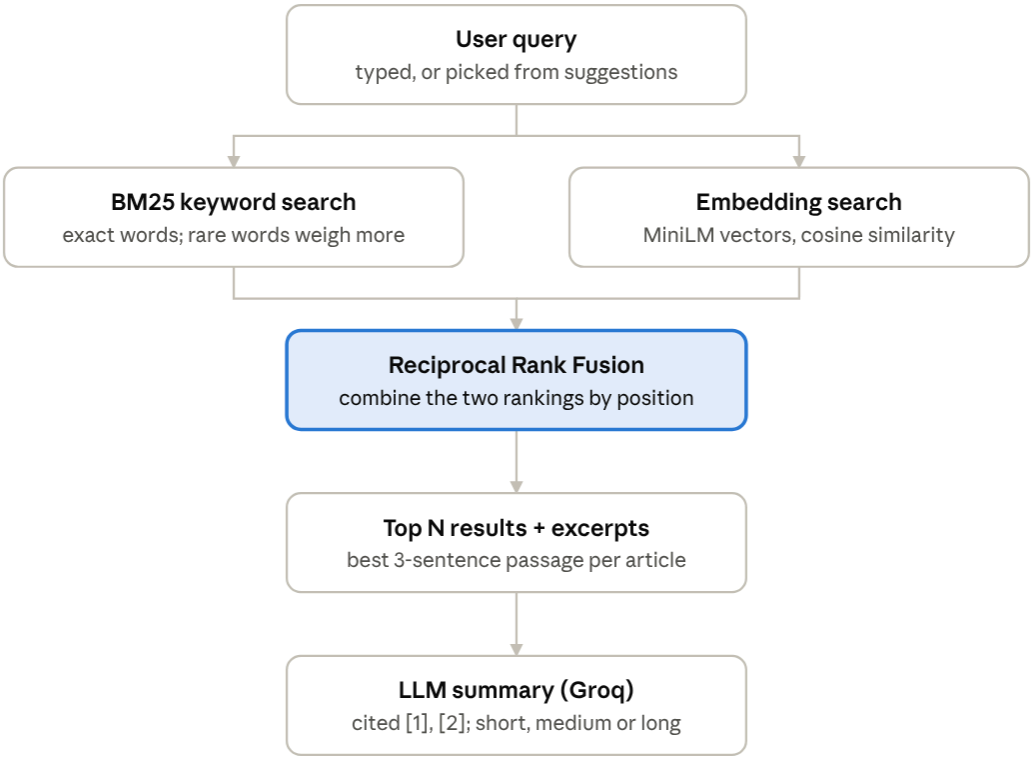
Step	What was done	Why
Load	Read each article and its summary (matched by file name) into one table	One table is easier to search and evaluate
Encoding	Read files as `latin-1`	Some files are not valid UTF-8
Title split	First line = title, rest = body	Titles are used for display and suggestions
Whitespace	Collapsed line breaks and extra spaces	Cleaner text for the LLM and excerpts
Joined sentences	`profits.The` → `profits. The`	Reference summaries had sentences stuck together
De-duplicate	Removed 136 duplicate or empty articles	Duplicates waste result slots and bias evaluation
Save	`doc_id` and `word_count` added; saved as `bbc_clean.csv`	Stable ids for evaluation; one clean file for all steps

No lowercasing or stop-word removal was applied to the stored text: LLMs and embeddings work best on natural text. Lowercasing is used only inside BM25.

Result: 2,089 unique articles, average 379 words (min 84, max 4,431).

2. Methodology

Two searches are fused by rank, then the LLM summarizes the top results



search and summarization pipeline · 6 steps

Search

- **BM25 (keyword search):** scores articles by query words, giving more weight to rare words and adjusting for article length. Chosen over TF-IDF because repeated words and long articles do not inflate the score.
- **Sentence embeddings (meaning search):** `all-MiniLM-L6-v2` turns each article into a vector; articles are ranked by cosine similarity to the query. Small, free, runs on a laptop; vectors are saved once to `embeddings.npy`.
- **Hybrid with Reciprocal Rank Fusion:** the two scores use different scales, so the rankings are combined by position: score = 1 / (60 + rank in BM25) + 1 / (60 + rank in embeddings). Articles ranked well in both lists rise to the top.
- **Excerpts:** each result's body is split into 3-sentence passages, and the one closest to the query is shown.

Why both methods: for *"people losing their jobs"*, BM25 returned off-topic word matches ("Cabs collect mountain of mobiles"), while embeddings found "Further rise in UK jobless total", which shares no keyword with the query.

Summarization

- **LLM:** `openai/gpt-oss-120b` through the Groq API, temperature 0.3. The top N articles (3–10) are sent with numbers and titles, long ones trimmed.
- **Prompt rules:** use only the given articles; write one connected summary; mark sources like [1]; skip irrelevant articles.
- **Length option:** short (~50 words), medium (~120), long (~250).

The "skip irrelevant" rule worked: for the jobs query, the summary ignored "US adds more jobs than expected", which search had ranked second.

3. Evaluation

Test set: 50 articles (10 per category, `random_state=42`). For each, the LLM wrote a 3–8 word search query without copying the title (e.g. "Fears raised over ballet future" → *decline in UK ballet students*). The correct answer is the article the query came from.

Search accuracy

Hit@1 / Hit@5 = share of queries with the correct article at rank 1 / in the top 5. MRR@10 = average of 1 / rank (0 if not in the top 10).

Method	Hit@1	Hit@5	MRR@10
BM25	0.82	0.98	0.895
Embeddings	0.80	0.98	0.884
Hybrid (RRF)	**0.82**	**1.00**	**0.907**

Hybrid matched or beat both methods and never missed the top 5. With 50 queries one query is 2 points, so the gaps are small. BM25 is helped by queries that contain exact names and numbers.

Summary quality: ROUGE

LLM summaries (medium) for the 50 test articles were compared with the reference summaries. Baseline: the article's first 3 sentences (Lead-3).

Metric (F1)	LLM	Lead-3
ROUGE-1	**0.453**	0.410
ROUGE-2	0.174	**0.327**

Metric (F1)	LLM	Lead-3
ROUGE-L	0.220	**0.330**

The LLM covers more key words (ROUGE-1) but loses on exact phrases, because the BBC reference summaries copy sentences from the article, as Lead-3 does, while the LLM rewrites them. ROUGE therefore under-rates rewritten summaries.

Summary quality: manual review

10 random summaries were rated 1–5 against the query and the full article by one reviewer.

Criterion	Score
Relevance	4.9
Coherence	4.1
Conciseness	3.7
Accuracy	3.6
Overall	**4.08**

Accuracy errors found: views credited to the wrong source, wording stronger than the article, cause-and-effect links the article does not make, and mixed-up numbers. ROUGE detects none of these, which is why the manual review was needed.

4. Challenges and solutions

Challenge	Solution
Dataset downloaded twice inside a nested folder	Kept one copy
136 duplicate articles; some near-duplicates remained	Removed exact duplicates; near-duplicates noted as a known issue
BM25 and embedding scores on different scales	Combined by rank with Reciprocal Rank Fusion
Windows memory error from loading the embedding model when it was not needed	Loaded search only where it is used
Excel saved the review sheet in Windows encoding	Read it with `cp1252`

Challenge	Solution
LLM wrote citations as `【5】`	Converted them to `[5]` with a regex
Prices like "$50 … $51" shown as a maths formula in Streamlit	Escaped `$` before display
Groq free tier: 8,000 tokens per minute caused 10–15 s hidden waits per summary	Trimmed articles from 800 to 400 words; lowered reasoning effort
Groq free tier: 200,000 tokens per day used up by evaluation runs	Spread runs over time; another model is a fallback
ROUGE favours copied text and misses factual errors	Added a Lead-3 baseline and a manual review